

3) a) $tr = 0.5s$

$$tr \approx 2 T_d \Rightarrow T_d = 0.25s \Rightarrow p = -4$$

b) $K = 3 \quad (G(0))$

$$p_1 = p_2 = -2$$

$$G(s) = \frac{A}{(s+2)^2} = \frac{A}{s^2 + 4s + 4}$$

$$G(0) = \frac{A}{4} = 3 \Rightarrow A = 12 \Rightarrow G(s) = \frac{12}{(s+2)^2}$$

4) a) i. $G(s) = \frac{s+2.7}{s(s+11)(s-15)}$

$$p_1 = 0 \quad p_2 = -11 \quad p_3 = 15 \rightarrow \text{нестабильн}$$

ii. $f(s) = s^5 + s^4 + 16s^3 + 9s^2 + 4s + 2$

$$\begin{array}{l|ll} s^5 & 1 & 16 & 4 \\ s^4 & 1 & 9 & 2 \\ s^3 & a & b & \\ s^2 & c & d & \\ s^1 & e & & \\ s^0 & f & & \end{array}$$

↑

сбв елементи у

Радіової колонки су

истої знака $\Rightarrow$ стабільн

$$a = \frac{1 \cdot 16 - 9 \cdot 1}{1} = 7$$

$$b = \frac{1 \cdot 4 - 2 \cdot 1}{1} = 2$$

$$c = \frac{a \cdot 9 - b \cdot 1}{a} = \frac{63 - 2}{7} = \frac{61}{7}$$

$$d = \frac{2a}{a} = 2$$

$$e = \frac{c \cdot b - a \cdot d}{c} = \frac{\frac{61}{7} \cdot 2 - 7 \cdot 2}{\frac{61}{7}} = 0.39$$

$$f = 2$$

$$b) W_p(s) = \frac{s+3a}{s^3+2s^2+(2a-b-1)s-a+2b}$$

$$f(s) = s^3+2s^2+(2a-b-1)s-a+2b + s+3a$$

$$f(s) = s^3+2s^2+(2a-b)s+2a+2b$$

$$\begin{array}{l|ll} s^3 & 1 & 2a-b \\ s^2 & 2 & 2a+2b \\ s^1 & c & \\ s^0 & 2a+2b & \end{array}$$

$$c = \frac{2(2a-b) - (2a+2b) \cdot 1}{2} = \frac{4a-2b-2a-2b}{2} = a-2b$$

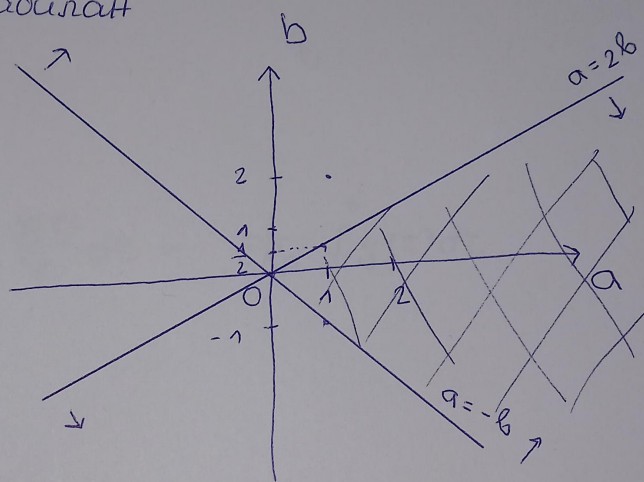
→ да да систем две стабилан

$$a-2b > 0$$

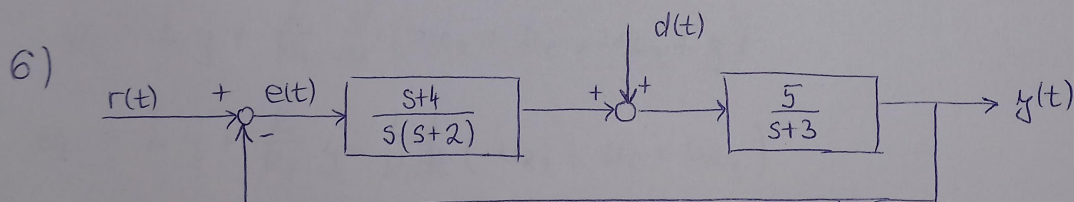
$$2a+2b > 0$$

$$a > 2b$$

$$a > -b$$



5) Канонске формуле не могу на 1. колоквијум!!!



$$r(t) = h(t) \xrightarrow{\mathcal{L}} R(s) = \frac{1}{s}$$

$$d(t) = 2th(t) \xrightarrow{\mathcal{L}} D(s) = \frac{2}{s^2}$$

$$W_{pp}(s) = \frac{5(s+4)}{s(s+2)(s+3)} \Rightarrow f(s) = s^3 + 5s^2 + 11s + 20$$

$$\begin{array}{l|ll} s^3 & 1 & 11 \\ s^2 & 5 & 20 \\ s^1 & 7 & \\ s^0 & 20 & \end{array} \Rightarrow \text{стабилан}$$

$$e_{ss} = e_{ssr} + e_{ssd}$$

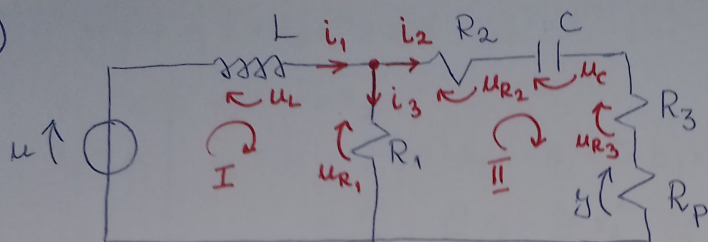
$e_{ssr} = 0 \Rightarrow$ имамо један асимптотички у директној линији

$$\frac{E(s)}{D(s)} = \frac{-5}{s+3} = \frac{-5s(s+2)}{s(s+2)(s+3)+5(s+4)}$$

$$e_{ssd} = \lim_{s \rightarrow 0} s \cdot \frac{-5s(s+2)}{s(s+2)(s+3)+5(s+4)} \cdot \frac{2}{s} = \frac{-10}{4} = -5$$

$$e_{ss} = 0 - 5 = -5$$

7)



$$u = u_L + u_{R1}$$

$$u_{R1} = u_{R2} + u_C + u_{R3} + u_{Rp}$$

$$i_2 = \frac{u}{R_p}$$

$$u_{R3} = i_2 \cdot R_3 = \frac{u}{R_p} R_3$$

$$u_C = \frac{1}{C} \int_0^t i_2(\tau) d\tau = \frac{1}{C} \int_0^t \frac{u(\tau)}{R_p} d\tau = \frac{1}{CR_p} \int_0^t u(\tau) d\tau$$

$$u_{R2} = i_2 \cdot R_2 = \frac{u}{R_p} R_2$$

$$u_L = L \frac{di_1(t)}{dt} = L \frac{d}{dt} (i_2 + i_3) = L \frac{d}{dt} \left(\frac{u}{R_p} + \frac{u_{R1}}{R_1} \right) = L \ddot{u} + \frac{L}{R_1} \frac{d}{dt} u_{R1}$$

$$u_L = L \ddot{u} + \frac{L}{R_1} \frac{d}{dt} (u_{R2} + u_C + u_{R3} + u_{Rp})$$

$$u_L = L \ddot{u} + \frac{L}{R_1} \dot{u} + \frac{L}{R_1} \frac{d}{dt} (u_{R2} + u_C + u_{R3})$$

$$u_L = L \ddot{u} + \frac{L}{R_1} \dot{u} + \frac{L}{R_1} \frac{d}{dt} \left(\frac{u}{R_p} R_2 + \frac{1}{CR_p} \int_0^t u d\tau + \frac{u}{R_p} R_3 \right)$$

$$u_L = L \ddot{u} + \frac{L}{R_1} \dot{u} + \frac{LR_2}{R_1 R_p} \dot{u} + \frac{L}{R_1 CR_p} u + \frac{LR_3}{R_1 R_p} \dot{u} = \frac{L}{R_1 CR_p} u + \dot{u} \left(L + \frac{L}{R_1} + \frac{LR_2}{R_1 R_p} + \frac{LR_3}{R_1 R_p} \right)$$

$$u = \frac{L}{R_1 CR_p} \dot{u} + \dot{u} \left(L + \frac{L}{R_1} + \frac{LR_2}{R_1 R_p} + \frac{LR_3}{R_1 R_p} \right) + \frac{R_2}{R_p} \dot{u} + \frac{1}{CR_p} \int_0^t u d\tau + \frac{R_3}{R_p} \dot{u}$$

$$8) \quad m \ddot{x} + b_1 \dot{x} + b_2 \dot{x}^2 + b_3 x + b_4 x^2 = F$$

$$F_0 = u_0 = 0$$

$$\begin{cases} x_1 = x \\ x_2 = \dot{x} \end{cases} \begin{cases} \dot{x}_1 = \dot{x} = x_2 \\ \dot{x}_2 = \ddot{x} = \frac{1}{m} (F - b_1 \dot{x} - b_2 \dot{x}^2 - b_3 x - b_4 x^2) \\ \quad = \frac{F}{m} - \frac{b_1}{m} x_2 - \frac{b_2}{m} x_2^2 - \frac{b_3}{m} x_1 - \frac{b_4}{m} x_1^2 \\ y = x_1 \end{cases}$$

$$a) \quad \dot{x}_1 \Big|_{x_{10}, x_{20}, F_0} = 0 \Rightarrow \boxed{x_{20} = 0}$$

$$\dot{x}_2 \Big|_{x_{10}, x_{20}, F_0} = 0 \Rightarrow \frac{F_0}{m} - \frac{b_1}{m} x_{20} - \frac{b_2}{m} x_{20}^2 - \frac{b_3}{m} x_{10} - \frac{b_4}{m} x_{10}^2 = 0$$

$$- x_{10} \left(\frac{b_3}{m} + x_{10} \frac{b_4}{m} \right) = 0$$

$$\boxed{x_{10} = 0} \vee \frac{b_3}{m} + x_{10} \frac{b_4}{m} = 0 \quad / \cdot m$$

$$MRT_1(0, 0, 0)$$

$$MRT_2\left(-\frac{b_3}{b_4}, 0, 0\right)$$

$$b_3 + x_{10} b_4 = 0$$

$$\boxed{x_{10} = -\frac{b_3}{b_4}}$$

$$b) \quad \Delta \dot{x}_1 = \Delta x_2$$

$$\Delta \dot{x}_2 = \left(-\frac{b_3}{m} - \frac{b_4}{m} 2x_{10}\right) \Delta x_1 + \left(-\frac{b_1}{m} - \frac{b_2}{m} 2x_{20}\right) \Delta x_2 + \frac{1}{m} \Delta F$$

$$\Delta y = \Delta x_1$$

$$MRT_1(0, 0, 0): \quad \Delta \dot{x}_1 = \Delta x_2$$

$$\Delta \dot{x}_2 = -\frac{b_3}{m} \Delta x_1 - \frac{b_1}{m} \Delta x_2 + \frac{1}{m} \Delta F$$

$$\Delta y = \Delta x_1$$

$$c) \quad \Delta \dot{x}_1 = \Delta x_2$$

$$\Delta \dot{x}_2 = -2\Delta x_1 - 3\Delta x_2 + \Delta F$$

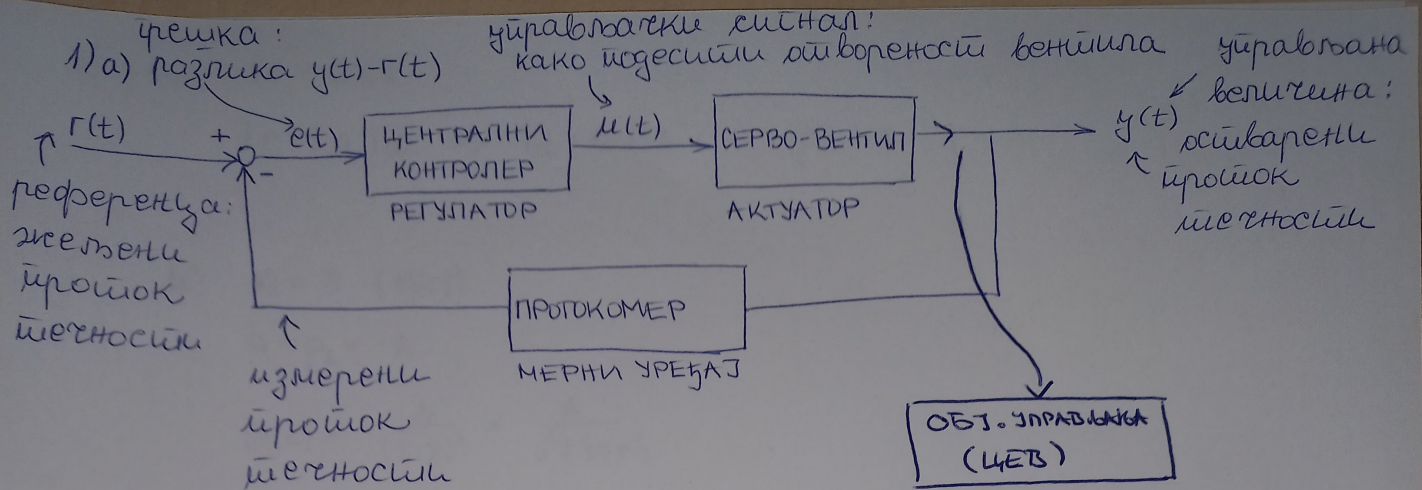
$$\Delta y = \Delta x_1$$

$$\Delta \dot{x} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \Delta x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Delta F$$

$$\Delta y = \begin{bmatrix} 1 & 0 \end{bmatrix} \Delta x$$

$$\det(sI - A) = \begin{vmatrix} s & 0 \\ 0 & s \end{vmatrix} - \begin{vmatrix} 0 & 1 \\ -2 & -3 \end{vmatrix} = \begin{vmatrix} s & -1 \\ 2 & s+3 \end{vmatrix} = s(s+3) + 2 = s^2 + 3s + 2 = (s+1)(s+2)$$

$$p_1 = -1 \quad p_2 = -2 \Rightarrow \text{стабильно}$$



б) $\rightarrow$ само линеарни, стационарни системи могу бити описани ф-јом преноса

$\rightarrow$ линеарни, стационарни системи задовољају принцип суперпозиције

$$2) \quad G(s) = \frac{3s+4}{(s+1)(s+3)}$$

$\rightarrow$ импулсни одзив $\Rightarrow u(t) = \delta(t) \xrightarrow{\mathcal{L}} U(s) = 1$

$$Y(s) = G(s) \cdot U(s) = \frac{3s+4}{(s+1)(s+3)} = \frac{A}{s+1} + \frac{B}{s+3}$$

$$A = \lim_{s \rightarrow -1} \left\{ (s+1) \frac{3s+4}{(s+1)(s+3)} \right\} = \frac{1}{2}$$

$$B = \lim_{s \rightarrow -3} \left\{ (s+3) \frac{3s+4}{(s+1)(s+3)} \right\} = \frac{5}{2}$$

$$Y(s) = \frac{1}{2} \frac{1}{s+1} + \frac{5}{2} \frac{1}{s+3}$$

$$y(t) = \mathcal{L}^{-1}\{Y(s)\} = \left(\frac{1}{2} e^{-t} + \frac{5}{2} e^{-3t} \right) h(t)$$